



Laser Authorized User Training

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Objectives

- Become an authorized user (able to use machines unsupervised)
- Learn normal operations procedures
- Learn safety procedures



Requirements

- Must have passed “Intro” course
- Must be paying member
- Must agree to follow policy documents and checklist



Policies

- You must carefully read, initial, and sign the policy document. This governs use of the machine and outlines your responsibility as operator.
- Copy provided by trainer
- Also available on GitHub, follow QR code on lasers.
- Policies
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General Provisions

- Provide your own materials
- Not under influence
- Log use and failures or safety issues
- Keep machine locked when not in use
- Follow items on checklist



Approved Materials

- If it's not on the approved materials list, don't cut it.
- If you disagree, get a copy of the Safety Data Sheet (SDS) and appeal to board for changes to policy, we can change if presented with a well-reasoned argument.
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Safety:

- Each type of safety condition should be handled depending on its nature.
 - Fire
 - Lens Vaporization
 - Tube Malfunction



Fire

- Most common
- Not a problem if handled quickly
- Big problem if left unattended
- Use E-Stop
- Extinguish fire, escalating as needed
 - Use fire blanket
 - Use extinguisher
 - Call 911
- Log failure and notify board.



Lens Vaporization

- Prevent by examining lens about monthly to make sure residue hasn't accumulated. If you don't want to clean it yourself, notify someone who can.
- Prevention is the most powerful tool here.
- Lenses cost ~\$40 and it's much easier, safer to replace a lens than to clean the machine.
- Keep lid closed
- Keep locked
- HAZMAT to avoid Selenium exposure
- Log failure and notify board
- Place sign on machine to warn of exposure.



Laser tube malfunction

- Tube can crack
 - Physical shock
 - Water freezing
- High-voltage risk
- E-stop
- Log failure, notify board member
- Keep locked, closed
- Sign on machine



Supervising others

- You may supervise general public WHO HAVE HAD INTRO TRAINING
 - We encourage this because it may help drive membership and donations
- You must watch them and stay nearby
- You are ultimately responsible for safety



Demonstrate Safe Operation

- We will now go to the lasers and
 - Prepare them for use
 - Unlock machine
 - Exhaust vents
 - Fan power
 - Air assist
 - Chillers



Demonstrate Safe Operation

- We will now go to the lasers and
 - Cut a small job
 - Focal length
 - Lightburn settings



Demonstrate Safe Operation

- We will now go to the lasers and
 - Close them down
 - Lock machine
 - Exhaust vents
 - Fan power
 - Air assist
 - Chillers



Examination

- Be prepared to answer questions about what to do in case of emergency
 - Fire
 - Lens vaporization
 - Tube malfunction



Congratulations, you are now authorized

- Signoff
- Authorized user badge/token

